复旦数学资料汇编
表示论专题

来源：香蕉空间 Solution 资料库

2026 年 10 月 13 日

编者说明

本册按课程汇集原站讲义、笔记、习题及试卷，保持原资料内容。格式转换不构成数学证明审核。每篇保留来源网址；原站未建立的页面不编造正文。内容采用原站标注的 CC BY-SA 4.0 许可，署名与详细历史可由原网页查看。

目录

第一章 表示论专题 **1**

1.1 笔记: 表示论专题 . . . 1
1.2 笔记: 表示论专题/2026.3.11 . . . 1
1.3 Recollection on f_*, f^*, $\otimes$, and $\mathcal{H}om(-,-)$ . . . 1
1.4 The Functor $f_!$ (!-pushforward) . . . 2
1.5 Proper Base Change . . . 4
1.6 The Functor $f^!$ (!-pullback) . . . 6
1.7 Verdier Duality Functor . . . 9
1.8 笔记: 表示论专题/2026.3.18 . . . 10
1.9 Open-closed distinguished triangle . . . 10
1.10 Smooth Morphisms . . . 14
1.11 Constructibility . . . 16
1.12 笔记: 表示论专题/2026.3.4 . . . 21
1.13 Set up for sheaf theory . . . 21
1.14 Local Systems . . . 21
1.15 Monodromy representations . . . 22
1.16 Constructible Sheaves . . . 24

第一章　表示论专题

1.1　笔记: 表示论专题

原网页

表示论专题 2026 春

（注：本内容尽可能忠实于上课的内容）

- 2026.3.4
- 2026.3.11
- 2026.3.18

1.2　笔记: 表示论专题/2026.3.11

原网页

1.3　Recollection on $f_*, f^*, \otimes$, and $\mathcal{H}om(-,-)$

定义 1.3.1. *Let X be a topological space, and let* $\mathrm{Sh}(X)$ *denote the category of sheaves of $\mathbb{C}$-vector spaces on X. The derived category of* $\mathrm{Sh}(X)$ *is defined as $D(X) = K(\mathrm{Sh}(X))[Q^{-1}]$. We define the associated bounded derived categories as follows:*

- $D^+(X) = \{\mathcal{F} \in D(X) \mid$ *there exists $N \in \mathbb{Z}$ such that $H^i(\mathcal{F}) = 0$ for all $i < N$*$\}$
- $D^-(X) = \{\mathcal{F} \in D(X) \mid$ *there exists $N \in \mathbb{Z}$ such that $H^i(\mathcal{F}) = 0$ for all $i > N$*$\}$
- $D^b(X) = \{\mathcal{F} \in D(X) \mid$ *there exists $N > 0$ such that $H^i(\mathcal{F}) = 0$ for all $|i| > N$*$\}$

For a continuous map $f : X \to Y$, the pullback functor $f^* : \mathrm{Sh}(Y) \to \mathrm{Sh}(X)$ is exact, whereas the pushforward functor $f_* : \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$ is left exact. This implies that there exist derived functors:

- $Rf^* = f^* : D^?(Y) \to D^?(X)$, where $? \in \{\emptyset, +, -, b\}$.
- $Rf_* : D^+(X) \to D^+(Y)$ (and respectively on D^b).

命题 1.3.1 (Adjunction)**.** *There is a natural adjunction isomorphism given by*

$$\mathrm{Hom}_{D(X)}(f^*\mathcal{G}, \mathcal{F}) \cong \mathrm{Hom}_{D(Y)}(\mathcal{G}, Rf_*\mathcal{F}).$$

定义 1.3.2. *For complexes $\mathcal{F},\mathcal{G}\in \mathrm{Ch}(X)$, we define their tensor product complex by*

$$(\mathcal{F}\otimes\mathcal{G})^n := \bigoplus_{p+q=n} \mathcal{F}^p\otimes\mathcal{G}^q,$$

equipped with the differential $d(x^p\otimes y^q)=(dx^p)\otimes y^q+(-1)^p x^p\otimes(dy^q)$.

This construction induces the left derived tensor product functor:

$$\otimes^L : D^-(X)\times D^-(X)\to D^-(X),\quad (\mathcal{F},\mathcal{G})\mapsto \mathcal{F}\otimes^L\mathcal{G}.$$

定义 1.3.3. *For complexes $\mathcal{F},\mathcal{G}\in\mathrm{Ch}(X)$, the global homomorphism complex $\mathrm{Hom}(\mathcal{F},\mathcal{G})$ belongs to $\mathrm{Ch}(pt)$. Its degree n component is given by*

$$\mathrm{Hom}^n(\mathcal{F},\mathcal{G})=\prod_{p\in\mathbb{Z}}\mathrm{Hom}(\mathcal{F}^p,\mathcal{G}^{p+n}),$$

with the corresponding differential defined as $d^n(\varphi)=d\circ\varphi+(-1)^{n+1}\varphi\circ d$.

Furthermore, we define the sheaf hom complex $\mathcal{H}om(\mathcal{F},\mathcal{G})\in\mathrm{Ch}(X)$ by associating to each open subset $U\subset X$ the complex $\mathrm{Hom}(\mathcal{F}|_U,\mathcal{G}|_U)$.

This sheafified version induces the right derived sheaf hom functor:

$$R\mathcal{H}om : D^-(X)^{\mathrm{op}}\times D^+(X)\to D^+(X).$$

命题 1.3.2 (Adjunction)**.** *For $\mathcal{F},\mathcal{G}\in D^-(X)$ and $\mathcal{A}\in D^+(X)$, we have the following adjunction isomorphism for derived functors:*

$$R\mathcal{H}om(\mathcal{F}\otimes^L\mathcal{G},\mathcal{A})\cong R\mathcal{H}om(\mathcal{F},R\mathcal{H}om(\mathcal{G},\mathcal{A})).$$

1.4 The Functor $f_!$ (!-pushforward)

定义 1.4.1. *A continuous map $f\colon X\to Y$ is **proper** if it is universally closed, that is, for any topological space Z, the product map $f\times\mathrm{id}_Z\colon X\times Z\to Y\times Z$ is closed.*

命题 1.4.1. *Let X and Y be Hausdorff, locally compact spaces, and let $f\colon X\to Y$ be a continuous map. Then the following are equivalent:*

1. *f is proper.*
2. *For any compact subset $K\subset Y$, the preimage $f^{-1}(K)\subset X$ is compact.*
3. *The map f is closed, and for every $y\in Y$, the fiber $f^{-1}(y)$ is compact.*

注 1.4.1. *For a finite type morphism $f\colon X\to Y$ between algebraic varieties, the induced continuous map on complex points $f\colon X(\mathbb{C})\to Y(\mathbb{C})$ is proper if and only if f is a proper morphism in the sense of algebraic geometry.*

定义 1.4.2. *For a continuous map $f\colon X\to Y$ and a sheaf $\mathcal{F}\in\mathrm{Sh}(X)$, we define the sheaf $f_!\mathcal{F}\in\mathrm{Sh}(Y)$ by specifying its sections over an open set $U\subset Y$:*

$$(f_!\mathcal{F})(U)=\{s\in\mathcal{F}(f^{-1}(U))\mid \text{the restricted map } f|_{\mathrm{supp}(s)}\colon \mathrm{supp}(s)\to U \text{ is proper}\}.$$

where $\mathrm{supp}(s)$ is by definition contained in $f^{-1}(U)$. Clearly, $f_!\mathcal{F}$ is a subsheaf of the pushforward sheaf $f_*\mathcal{F}$.

引理 1.4.1. *The functor $f_!\colon \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$ is left exact.*

Hint: This follows from the fact that $f_!\mathcal{F}$ is defined as a subsheaf of $f_*\mathcal{F}$ and the properties of its definition. □

As a consequence of this left exactness, there exists a right derived functor $Rf_!\colon D^+(X) \to D^+(Y)$.

Convention*: *Whenever the functor $f_!$ is involved, we will assume that all spaces are locally compact.

For the constant map to a point $a_X\colon X \to \mathrm{pt}$, we denote the space of compactly supported sections by

$$\Gamma_c(\mathcal{F}) = a_{X!}\mathcal{F} = \{s \in \Gamma(\mathcal{F}) \mid \mathrm{supp}(s) \text{ is compact}\}.$$

This induces the derived functor $R\Gamma_c := Ra_{X!}$, and we define **the cohomology with compact support** as

$$H_c^k(\mathcal{F}) := H^k(R\Gamma_c(\mathcal{F})). \tag{1.1}$$

例 1.4.1. *If a continuous map $f\colon X \to Y$ is proper, then $f_! = f_*$.*

Proof. It suffices to show that for all $s \in \mathcal{F}(f^{-1}(U))$, the map $\mathrm{supp}(s) \to U$ is proper. Note that an immersion $Z \hookrightarrow U$ is proper if and only if Z is closed in U.

Consider the sequence of maps:

$$\mathrm{supp}(s) \hookrightarrow f^{-1}(U) \xrightarrow{f} U.$$

By definition, $\mathrm{supp}(s)$ is closed in $f^{-1}(U)$, so the inclusion $\mathrm{supp}(s) \hookrightarrow f^{-1}(U)$ is proper. Since $f\colon X \to Y$ is proper, its base change $f^{-1}(U) \to U$ is also proper. Therefore, the composition $\mathrm{supp}(s) \to U$ is proper, which completes the proof. □

例 1.4.2 (Extension by zero)**.** *Let $j\colon Y \to X$ be an embedding of a locally closed subspace, and let $\mathcal{F} \in \mathrm{Sh}(Y)$. Then the stalks of $j_!\mathcal{F}$ are given by*

$$(j_!\mathcal{F})_x = \begin{cases} \mathcal{F}_x, & \text{if } x \in Y, \\ 0, & \text{if } x \notin Y. \end{cases}$$

In particular, the functor $j_!$ is exact.

Moreover, $j_!\mathcal{F}$ is the sheafification of the presheaf $j_{!,pre}\mathcal{F}$ defined by

$$j_{!,pre}\mathcal{F}(U) = \begin{cases} \mathcal{F}(U \cap Y), & \text{if } U \cap \overline{Y} \subset Y, \\ 0, & \text{otherwise}. \end{cases}$$

Proof. Assume $Y = Y_1 \cap Y_2$, where Y_1 is an open subset and Y_2 is a closed subset of X.

Case 1: If $x \in Y$, we will show that $(j_!\mathcal{F})_x = \mathcal{F}_x$.

Take an open neighborhood $U \subseteq Y_1$ such that $x \in U$. Then $U \cap Y = U \cap Y_1 \cap Y_2 = U \cap Y_2$. For any section $s \in \mathcal{F}(U \cap Y)$, we have the inclusion $\mathrm{supp}(s) \hookrightarrow U \cap Y = U \cap Y_2$.

Since Y_2 is closed in X, $U \cap Y_2$ is closed in U. Thus $\mathrm{supp}(s)$, being closed in $U \cap Y$, is also closed in U. This implies that the map $\mathrm{supp}(s) \to U$ is proper. Therefore, every section in $\mathcal{F}(U \cap Y)$ has proper support over U, which means $(j_!\mathcal{F})(U) = \mathcal{F}(U \cap Y)$. Passing to the stalk at x yields $(j_!\mathcal{F})_x = \mathcal{F}_x$.

Case 2: If $x \notin Y$.

Let $U \subseteq X$ be an open neighborhood of x. For any section $s \in (j_!\mathcal{F})(U) \subseteq \mathcal{F}(U \cap Y)$, the map $\mathrm{supp}(s) \hookrightarrow U \cap Y \hookrightarrow U$ is proper. This properness implies that $\mathrm{supp}(s)$ is closed in U.

Since $s \in \mathcal{F}(U \cap Y)$, we know $\mathrm{supp}(s) \subseteq Y$. Because $x \notin Y$, it follows that $x \notin \mathrm{supp}(s)$. Therefore, $x \in U \setminus \mathrm{supp}(s)$. Since $\mathrm{supp}(s)$ is closed in U, its complement $U \setminus \mathrm{supp}(s)$ is an open neighborhood of x.

Restricting s to this smaller neighborhood, we obtain $s|_{U \setminus \mathrm{supp}(s)} = 0$. Thus, the germ of s at x is 0, which implies $(j_!\mathcal{F})_x = 0$.

For the presheaf $j_{!,\mathrm{pre}}\mathcal{F}$, consider an open subset $U \subset X$ such that $U \cap \overline{Y} \subset Y$. Any closed subset of $U \cap Y$ is also closed in U. This holds because if $V \subset X$ is a closed subset, then $U \cap Y \cap V = U \cap \overline{Y} \cap V$ by our given condition. Thus, we have

$$(j_!\mathcal{F})(U) = \{s \in \mathcal{F}(U \cap Y) \mid \text{the map } \mathrm{supp}(s)(\hookrightarrow U \cap Y \hookrightarrow) U \text{ is proper}\} = \mathcal{F}(U \cap Y).$$

This implies that there is a natural morphism from the sheafification $(j_{!,\mathrm{pre}}\mathcal{F})^+$ to $j_!\mathcal{F}$.

To prove that this morphism is an isomorphism, we need to show that the stalks are given by

$$(j_{!,\mathrm{pre}}\mathcal{F})_x = \begin{cases} \mathcal{F}_x, & \text{if } x \in Y, \\ 0, & \text{if } x \notin Y. \end{cases}$$

We verify this case by case:

- At $x \notin \overline{Y}$, it is clear that $(j_{!,\mathrm{pre}}\mathcal{F})_x = 0$.
- At $x \in \overline{Y} \setminus Y$, every open neighborhood U of x satisfies $U \cap \overline{Y} \not\subset Y$. Therefore, the stalk is zero, i.e., $(j_{!,\mathrm{pre}}\mathcal{F})_x = 0$.
- At $x \in Y$, since Y is a locally closed subspace, we can write $Y = Y_1 \cap Y_2$, where Y_1 is open and Y_2 is closed. Let U be an open neighborhood of x such that $x \in U \subset Y_1$. Then $U \cap \overline{Y} \subset Y_1 \cap \overline{Y_2} \subset Y_1 \cap Y_2 = Y$. This implies that $(j_{!,\mathrm{pre}}\mathcal{F})_x = \mathcal{F}_x$.

This completes the proof. □

Notation and Conventions: Henceforth, we will use f_*, f^*, and $f_!$ to denote the corresponding functors for derived categories. For the corresponding functors in abelian categories, we will use the notation $^\circ f_*$, $^\circ f^*$, and $^\circ f_!$.

Furthermore, for the composition of morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$, we have the following natural isomorphisms:

$$(g \circ f)^* \simeq f^* \circ g^*, \quad (g \circ f)_* \simeq g_* \circ f_*, \quad \text{and} \quad (g \circ f)_! \simeq g_! \circ f_!.$$

1.5 Proper Base Change

Let us consider a Cartesian diagram of topological spaces:

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[g]{} & Y \end{array}$$

定理 1.5.1. *For any $\mathcal{F} \in D^+(X)$, there exists a natural isomorphism*

$$g^* f_! \mathcal{F} \xrightarrow{\sim} f'_! (g')^* \mathcal{F}.$$

注 1.5.1. *If f is proper, then $f_! = f_*$, which means the "$!$" can be replaced by "$*$" on the right-hand side. In general, this is not true. This theorem is very useful.*

Before proving the theorem, we first establish the following lemma.

引理 1.5.1. *For any $\mathcal{G} \in D^+(X')$, there exists a natural morphism*

$$f_! g'_* \mathcal{G} \longrightarrow g_* f'_! \mathcal{G}.$$

Proof Sketch of Lemma. Let $h = f \circ g' = g \circ f' \colon X' \to Y$. To establish the morphism, we compare the sections over an open subset $U \subset Y$ at the level of abelian categories.

For the left-hand side, we have:

$$(^\circ f_! \, ^\circ g'_* \mathcal{G})(U) = \{s \in (^\circ g'_* \mathcal{G})(f^{-1}(U)) \mid \text{the map } f|_{\mathrm{supp}_X(s)} \colon \mathrm{supp}_X(s) \to U \text{ is proper}\}$$

For the right-hand side, we have:

$$\begin{aligned}(^\circ g_* \, ^\circ f'_! \mathcal{G})(U) &= (^\circ f'_! \mathcal{G})(g^{-1}(U)) \\ &= \{s \in \mathcal{G}((f')^{-1}(g^{-1}(U))) \mid \text{the map } f'|_{\mathrm{supp}_{X'}(s)} \colon \mathrm{supp}_{X'}(s) \to g^{-1}(U) \text{ is proper}\}\end{aligned}$$

Notice that the underlying group of sections is the same, since

$$(^\circ g'_* \mathcal{G})(f^{-1}(U)) = \mathcal{G}((g')^{-1}(f^{-1}(U))) = \mathcal{G}(h^{-1}(U)) = \mathcal{G}((f')^{-1}(g^{-1}(U))).$$

Now we compare the properness conditions. We have the following diagram of supports:

$$\begin{array}{ccccc} \mathrm{supp}_{X'}(s) & \hookrightarrow & (g')^{-1}(\mathrm{supp}_X(s)) & \longrightarrow & \mathrm{supp}_X(s) \\ & \searrow^{\text{proper}} & \downarrow{\scriptstyle \text{proper}} & & \downarrow{\scriptstyle \text{proper}} \\ & & g^{-1}(U) & \longrightarrow & U \end{array}$$

If the map $\mathrm{supp}_X(s) \to U$ is proper, then by the property of Cartesian squares, its base change $(g')^{-1}(\mathrm{supp}_X(s)) \to g^{-1}(U)$ is also proper. Since $\mathrm{supp}_{X'}(s)$ is a closed subset of $(g')^{-1}(\mathrm{supp}_X(s))$, the restriction $\mathrm{supp}_{X'}(s) \to g^{-1}(U)$ is also proper.

Therefore, any section satisfying the properness condition for the left-hand side also satisfies the properness condition for the right-hand side. This gives an inclusion:

$$(^\circ f_! \, ^\circ g'_* \mathcal{G})(U) \subseteq (^\circ g_* \, ^\circ f'_! \mathcal{G})(U).$$

To prove the lemma, it is sufficient to check this inclusion in the abelian category case. This completes the proof of the lemma. □

Proof Sketch of Theorem. Let $\mathcal{G} = (g')^*\mathcal{F}$. We have the following sequence of morphisms:

$$g^* f_! \mathcal{F} \longrightarrow g^* f_! g'_* (g')^* \mathcal{F} \xrightarrow{\text{by Lemma}} g^* g_* f'_! (g')^* \mathcal{F} \longrightarrow f'_! (g')^* \mathcal{F}.$$

To show that this composition is an isomorphism, one checks it on the stalks. The detailed proof is omitted here.

(Hint: This can be proved by reduction to the case where Y' is a point. For details, see [Kashiwara-Shapira, Sheaves on Manifolds]). □

1.6 The Functor $f^!$ (!-pullback)

定义 1.6.1. *Let $f\colon X \to Y$ be a continuous map. The functor $^\circ f_!\colon \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$ is said to **have finite cohomological dimension** if there exists an integer $r \in \mathbb{N}$ such that for all $k > r$ and for all sheaves $\mathcal{F} \in \mathrm{Sh}(X)$, we have $R^k(^\circ f_!)\mathcal{F} = 0$.*

This condition is denoted by $\dim(^\circ f_!) < +\infty$.

例 1.6.1. *Let X be an algebraic or analytic variety. Every sheaf $\mathcal{F} \in \mathrm{Sh}(X)$ has an injective resolution of length at most $2\dim X$. This implies that for any continuous map $f\colon X \to Y$ between varieties, we always have* $\dim(^\circ f_!) < +\infty$.

定理 1.6.1 (Verdier Duality)**.** *Let $f\colon X \to Y$ be a continuous map such that* $\dim(^\circ f_!) < +\infty$. *Then there exists a triangulated functor $f^!\colon D^+(Y) \to D^+(X)$ such that we have a natural isomorphism*

$$R\mathcal{H}om(f_!\mathcal{F}, \mathcal{G}) \simeq f_* R\mathcal{H}om(\mathcal{F}, f^!\mathcal{G})$$

for all $\mathcal{F} \in D^-(X)$ and all $\mathcal{G} \in D^+(Y)$. In other words, the functor $f_!$ has a right adjoint.

注 1.6.1. *A sheaf $\mathcal{F} \in \mathrm{Sh}(X)$ is **f-soft** if for every fiber X_y of f, $\mathcal{F}|_{X_y}$ is c-soft. A sheaf $\mathcal{F}$ is **c-soft** if for every compact subset $K \subset X$, the restriction map $\Gamma(X, \mathcal{F}) \to \Gamma(K, \mathcal{F}|_K)$ is surjective.*

Proof Sketch. Fix a finite f-soft resolution $\mathcal{K}$ of the constant sheaf $\mathbb{C}_X$. Such a resolution exists by the assumption that $^\circ f_!$ has finite cohomological dimension.

For any complex $\mathcal{G} \in \mathrm{Ch}^+(\mathrm{Sh}(Y))$, we define a presheaf $f^!\mathcal{G}$ as follows. For each open subset $U \subset X$ with the inclusion $j_U\colon U \hookrightarrow X$, we set

$$(f^!\mathcal{G})(U) := \mathrm{Hom}(^\circ f_! j_{U!}(\mathcal{K}|_U), \mathcal{G}).$$

One then verifies the following properties:

- $f^!\mathcal{G}$ is a complex of sheaves.
- It induces a well-defined functor $D^+(Y) \to D^+(X)$.
- There is a natural isomorphism $R\mathcal{H}om(f_!\mathcal{F}, \mathcal{G}) \simeq f_* R\mathcal{H}om(\mathcal{F}, f^!\mathcal{G})$.

This isomorphism is clear, for example, when $\mathcal{F} = \mathbb{C}_X$. For detailed verifications, we refer to [Kashiwara-Shapira, §3.1]. □

By the uniqueness of right adjoints, we have a natural isomorphism $(g \circ f)^! \simeq f^! \circ g^!$.

注 1.6.2. *In general, $f^!$ is not the derived functor of its zeroth cohomology, that is, $f^! \neq R(^\circ f^!)$. However, this is true in some special cases.*

例 1.6.2 (Restriction with supports in X)**.** *Let $f\colon X \hookrightarrow Y$ be an inclusion of a locally closed subspace. We define the functor $^\circ f^!\colon \mathrm{Sh}(Y) \to \mathrm{Sh}(X)$ at the level of abelian categories as follows. For any open subset $U \subset X$ and any sheaf $\mathcal{F} \in \mathrm{Sh}(Y)$, we set*

$$(^\circ f^!\mathcal{F})(U) = \varinjlim_{\substack{V \subseteq Y \text{ open} \\ V \cap \overline{X} = U}} \{s \in \mathcal{F}(V) \mid \mathrm{supp}(s) \subseteq U\}.$$

(Homework: verify that $^\circ f^!\mathcal{F}$ is indeed a sheaf on X).

In particular, for an open subset X, we have $^\circ f^!\mathcal{F} = {}^\circ f^*\mathcal{F}$. In general, there is a natural morphism $^\circ f^! \longrightarrow {}^\circ f^*$.

One can check that the functor $^\circ f^!$ is left exact. Thus, we can define its right derived functor as

$$f^! = R(^\circ f^!)\colon D^+(Y) \longrightarrow D^+(X).$$

命题 1.6.1. *There is a natural adjunction isomorphism:*

$$\mathrm{Hom}_{D(Y)}(f_!\mathcal{F}, \mathcal{G}) \simeq \mathrm{Hom}_{D(X)}(\mathcal{F}, f^!\mathcal{G}).$$

注 1.6.3. *By the uniqueness of right adjoints, this derived functor coincides with the functor $f^!$ defined previously.*

Proof Sketch. Recall that the presheaf pushforward with proper supports is given by

$$(f_{!,\mathrm{pre}}\mathcal{F})(U) = \begin{cases} \mathcal{F}(U \cap X), & \text{if } U \cap \overline{X} \subseteq X, \\ 0, & \text{otherwise.} \end{cases}$$

To establish the adjunction, we describe the unit and counit morphisms.

For the unit $\mathcal{F} \longrightarrow {}^\circ f^! f_{!,\mathrm{pre}}\mathcal{F}$: given an open subset $U \subseteq X$ and an open subset $V \subseteq Y$ such that $V \cap \overline{X} = U$, we clearly have $U = V \cap X$. The morphism is defined by the identity map

$$\mathcal{F}(U) \longrightarrow \mathcal{F}(V \cap X) = \mathcal{F}(U),$$

For the counit $f_{!,\mathrm{pre}}{}^\circ f^!\mathcal{G} \longrightarrow \mathcal{G}$: for an open subset $V \subseteq Y$ with $V \cap \overline{X} \subseteq X$, where $U = V \cap X = V \cap \overline{X}$. The morphism is given by the natural inclusion

$$\{s \in \mathcal{G}(V) \mid \mathrm{supp}(s) \subseteq U\} \longrightarrow \mathcal{G}(V).$$

This completes the proof. □

定义 1.6.2. *Let $i_x\colon \{x\} \hookrightarrow X$ be the inclusion of a point. The complex $i_x^!\mathcal{F}$ is called the **costalk** of $\mathcal{F}$ at x.*

定义 1.6.3. *For the constant map to a point $a_X\colon X \longrightarrow pt$, assume that the cohomological dimension satisfies $\dim(^\circ a_{X!}) < +\infty$. We define the **dualizing sheaf** of X as*

$$\omega_X := a_X^!\mathbb{C} \in D^+(X).$$

引理 1.6.1. *The i-th cohomology sheaf $\mathcal{H}^i(\omega_X) \in \mathrm{Sh}(X)$ is the sheaf associated to the presheaf defined by*

$$V \longmapsto (H_c^{-i}(V))^\vee,$$

where the right-hand side represents the dual space of the compactly supported cohomology in degree $-i$.

注 1.6.4. *We define $H_c^k(V) := H_c^k(\mathbb{C}_V)$ which is defined in (1.1).*

Proof Sketch. Recall that for a complex $\mathcal{F} \in D^+(X)$, its i-th cohomology sheaf $\mathcal{H}^i(\mathcal{F})$ is the sheaf associated to the presheaf given by

$$U \longmapsto \mathrm{Hom}_{D(X)}(\mathbb{C}_U, \mathcal{F}[i]).$$

For the dualizing sheaf $\mathcal{F} = \omega_X = a_X^!\mathbb{C}$, we can compute the sections of this presheaf as follows:

$$\mathrm{Hom}_{D(U)}(\mathbb{C}_U, a_U^!\mathbb{C}[i]) \simeq \mathrm{Hom}_{D(\mathrm{pt})}(a_{U!}\mathbb{C}_U[i], \mathbb{C}) = (H^{-i}(R\Gamma_c(\mathbb{C}_U)))^\vee = (H_c^{-i}(U))^\vee.$$

This establishes the lemma. □

注 1.6.5.

1. *The cohomology $H^{-i}(X, \omega_X)$ is defined as the Borel-Moore homology $H_i^{BM}(X)$. This is particularly useful in geometric representation theory.*
2. *Next time, we will discuss that for a smooth morphism f between varieties of relative complex dimension d, we have the isomorphism*

   $$f^!\mathcal{F} \simeq f^*\mathcal{F}[2d].$$

   (Note: the shift (d) is often called the "Tate twist" in related contexts).
   In particular, for a smooth variety X, the dualizing sheaf is given by $\omega_X = \mathbb{C}_X[2\dim_{\mathbb{C}} X]$.
   Furthermore, for $d > 0$, we have $\mathcal{H}^0(f^!) = \mathcal{H}^0(f^*[2d]) = 0$ because the pullback functor f^* is exact. However, the cohomology at other degrees $\mathcal{H}^i$ is not necessarily zero. Therefore, we can see that $R(\mathcal{H}^0(f^!)) = 0$, which is clearly not isomorphic to $f^!$.
   More generally, for an orientable real manifold X of real dimension n, the dualizing sheaf is isomorphic to $\mathbb{C}_X[n]$. Consider the adjunction for the constant map $a_X \colon X \to pt$:

   $$\begin{aligned} R\mathcal{H}om(a_{X!}\mathbb{C}_X, \mathbb{C}) &\simeq a_{X*}R\mathcal{H}om(\mathbb{C}_X, a_X^!\mathbb{C}) \\ &= a_{X*}R\mathcal{H}om(\mathbb{C}_X, \mathbb{C}_X[n]) \\ &= R\Gamma(X)[n]. \end{aligned}$$

   On the other hand, the left-hand side can be computed directly as:

   $$LHS = (R\Gamma_c(X))^\vee.$$

   By taking the $-k$-th cohomology H^{-k} on both sides of the isomorphism $(R\Gamma_c(X))^\vee \simeq R\Gamma(X)[n]$, we obtain

   $$(H_c^k(X))^\vee \simeq H^{n-k}(X),$$

   which is exactly the statement of Poincaré duality.
3. *For a continuous map $f \colon X \to Y$, we have the relation $f^!\omega_Y \simeq \omega_X$.*

1.7 Verdier Duality Functor

定义 1.7.1. *The **Verdier duality functor** is defined as:*

$$\mathbb{D} = \mathbb{D}_X \colon D^-(X)^{op} \longrightarrow D^+(X),$$

where the action on a complex $\mathcal{F}$ is given by

$$\mathbb{D}(\mathcal{F}) = R\mathcal{H}om(\mathcal{F}, \omega_X).$$

引理 1.7.1. *For a continuous map $f \colon X \to Y$, we have the following natural isomorphisms:*

1. $f_* \mathbb{D}\mathcal{F} \simeq \mathbb{D}(f_! \mathcal{F})$
2. $f^! \mathbb{D}\mathcal{F} \simeq \mathbb{D}(f^* \mathcal{F})$

Proof. Recall the Verdier duality adjunction for a map $f \colon X \to Y$:

$$R\mathcal{H}om(f_!\mathcal{F}, \mathcal{G}) \simeq f_* R\mathcal{H}om(\mathcal{F}, f^!\mathcal{G}).$$

(1) To prove the first isomorphism, we let $\mathcal{G} = \omega_Y$. The left-hand side becomes $R\mathcal{H}om(f_!\mathcal{F}, \omega_Y) = \mathbb{D}(f_!\mathcal{F})$. The right-hand side becomes $f_* R\mathcal{H}om(\mathcal{F}, f^!\omega_Y)$. Since $f^!\omega_Y \simeq \omega_X$, this is exactly $f_*\mathbb{D}\mathcal{F}$.

(2) To prove the second isomorphism, we first need the following theorem:

定理 1.7.1 (Projection Formula)**.** *There is a natural isomorphism:*

$$f_!\mathcal{F} \otimes^L \mathcal{G} \simeq f_!(\mathcal{F} \otimes^L f^*\mathcal{G}). \tag{1.2}$$

The detailed proof can be found in [Kashiwara-Shapira, Proposition 2.6.6].

From the projection formula, we can deduce the following fundamental isomorphism:

$$f^! R\mathcal{H}om(\mathcal{F}_1, \mathcal{F}_2) \simeq R\mathcal{H}om(f^*\mathcal{F}_1, f^!\mathcal{F}_2). \tag{1.3}$$

In fact, for any complex $\mathcal{F}$, we have the following sequence of natural isomorphisms:

$$\begin{aligned}
\mathrm{Hom}(\mathcal{F}, f^! R\mathcal{H}om(\mathcal{F}_1, \mathcal{F}_2)) &\simeq \mathrm{Hom}(f_!\mathcal{F} \otimes^L \mathcal{F}_1, \mathcal{F}_2) \\
&\simeq \mathrm{Hom}(f_!(\mathcal{F} \otimes^L f^*\mathcal{F}_1), \mathcal{F}_2) \\
&\simeq \mathrm{Hom}(\mathcal{F} \otimes^L f^*\mathcal{F}_1, f^!\mathcal{F}_2) \\
&\simeq \mathrm{Hom}(\mathcal{F}, R\mathcal{H}om(f^*\mathcal{F}_1, f^!\mathcal{F}_2)).
\end{aligned}$$

Since this holds for any $\mathcal{F}$, the isomorphism (1.3) follows.

Finally, let us substitute $\mathcal{F}_1 = \mathcal{F}$ and $\mathcal{F}_2 = \omega_Y$ into (1.3). This yields:

$$f^! R\mathcal{H}om(\mathcal{F}, \omega_Y) \simeq R\mathcal{H}om(f^*\mathcal{F}, f^!\omega_Y).$$

Notice that the left-hand side is $f^!\mathbb{D}\mathcal{F}$. For the right-hand side, since $f^!\omega_Y \simeq \omega_X$, it becomes $R\mathcal{H}om(f^*\mathcal{F}, \omega_X) = \mathbb{D}(f^*\mathcal{F})$. This completes the proof. □

注 1.7.1. *If we expect the Verdier duality functor to be an involution, that is,* $\mathbb{D}^2 = \mathrm{Id}$*, we need to restrict our attention to complexes* $\mathcal{F} \in D_c^b(X)$*. This implies that we need the fact that all functors above keep constructible sheaves.*

By the properties of the duality functor, it interchanges the pushforward and pullback operations. This yields the following relations, which are extremely useful in computations:

$$f_! = \mathbb{D} \circ f_* \circ \mathbb{D}, \tag{1.4}$$

$$f^! = \mathbb{D} \circ f^* \circ \mathbb{D}. \tag{1.5}$$

An important geometric case arises when we consider a closed subspace Z of X and its open complement $U = X \setminus Z$. Let i and j be the corresponding closed and open immersions, respectively:

$$Z \overset{i}{\hookrightarrow} X \overset{j}{\hookleftarrow} U.$$

For any complex $\mathcal{F}$, there exist the following two fundamental distinguished triangles:

$$j_! j^* \mathcal{F} \longrightarrow \mathcal{F} \longrightarrow i_* i^* \mathcal{F} \xrightarrow{+1}$$

and

$$i_* i^! \mathcal{F} \longrightarrow \mathcal{F} \longrightarrow j_* j^* \mathcal{F} \xrightarrow{+1}.$$

the second triangle can be “obtained” by applying Verdier functor and (1.4) to the first one.

1.8　笔记: 表示论专题/2026.3.18

原网页

注 1.8.1. *From now on, for* $\mathcal{F} \in D(X)$*, the notation* $H^k(\mathcal{F})$ *and* $\mathcal{H}^k(\mathcal{F})$ *will be used interchangeably to denote the* k*-th cohomology sheaf of* $\mathcal{F}$*.*

1.9　Open-closed distinguished triangle

Let $i\colon Z \hookrightarrow X$ be a closed subset, and let $j\colon U = X \setminus Z \hookrightarrow X$ be the open complement. We have the relations:

$$i_* = i_! \quad \text{and} \quad j^* = j^!$$

定理 1.9.1. *For the inclusions* $i\colon Z \hookrightarrow X$ *and* $j\colon U \hookrightarrow X$*, the following hold:*

1. $i^* j_! = i^! j_* = j^* i_* = 0$.
2. *For any sheaf* $\mathcal{F} \in D(X)$*, there exists a natural distinguished triangle:*

$$j_! j^* \mathcal{F} \longrightarrow \mathcal{F} \longrightarrow i_* i^* \mathcal{F} \xrightarrow{+1}$$

3. *For any sheaf* $\mathcal{F} \in D(X)$*, there exists a natural distinguished triangle:*

$$i_* i^! \mathcal{F} \longrightarrow \mathcal{F} \longrightarrow j_* j^* \mathcal{F} \xrightarrow{+1}$$

注 1.9.1. *Notice that the first maps in the triangles essentially use "$!$", while the second maps use "$*$". This pattern works because $i_* = i_!$ and $j^* = j^!$.*

Proof. It suffices to show these properties in the abelian category of sheaves.

For (1), consider the stalks of the sheaves. For a sheaf $\mathcal{F}$, the stalk $(j_!\mathcal{F})_x = 0$ for $x \notin U$. Similarly, $(i_!\mathcal{F})_x = 0$ for $x \notin Z$. This immediately implies that:

$$i^* j_! = j^* i_* = 0$$

By Verdier duality, the functor $i^! \circ j_*$ is the right adjoint to the functor $j^* \circ i_* = 0$. Because the right adjoint of the zero functor is also zero, it follows that $i^! \circ j_* = 0$.

For (2), we utilize the properties of adjunction. From the adjunction isomorphism $\mathrm{Hom}(j^*\mathcal{F}, j^*\mathcal{F}) \cong \mathrm{Hom}(j_!j^*\mathcal{F}, \mathcal{F})$, the identity map on the left induces a natural counit map:

$$j_!j^*\mathcal{F} \longrightarrow \mathcal{F}$$

The natural unit map $\mathcal{F} \longrightarrow i_*i^*\mathcal{F}$ can also be obtained in a similar manner.

In abelian categories, for a sheaf $\mathcal{F} \in \mathrm{Sh}(X)$, we can verify exactness by checking the stalks. Consider the sequence of stalks:

$$(j_!j^*\mathcal{F})_x \longrightarrow \mathcal{F}_x \longrightarrow (i_*i^*\mathcal{F})_x \tag{*}$$

For $x \in U$, we have $(j_!j^*\mathcal{F})_x = \mathcal{F}_x$ and $(i_*i^*\mathcal{F})_x = 0$. For $x \notin U$, we have $(j_!j^*\mathcal{F})_x = 0$ and $(i_*i^*\mathcal{F})_x = \mathcal{F}_x$. In both cases, the sequence (*) is exact. This exact sequence in the abelian category induces the corresponding distinguished triangle in the derived category.

For (3), first we consider the abelian category. By definition, for any open set V, the sections of the pushforward of the pullback are given by:

$$(j_*j^*\mathcal{F})(V) = \mathcal{F}(V \cap U)$$

The kernel of the unit map $\mathcal{F} \longrightarrow j_*j^*\mathcal{F}$ evaluated on V is:

$$\begin{aligned}\ker(\mathcal{F} \longrightarrow j_*j^*\mathcal{F})(V) &= \{s \in \mathcal{F}(V) \mid s|_{V\cap U} = 0\} \\ &= \{s \in \mathcal{F}(V) \mid \operatorname{supp} s \subseteq V \cap Z\}\end{aligned}$$

Since Z is a closed subset (i.e., $Z = \overline{Z}$), this coincides with the sections with supports in Z, namely $(i^!\mathcal{F})(V \cap Z)$. Therefore, the following sequence is exact:

$$0 \longrightarrow i_*i^!\mathcal{F} \longrightarrow \mathcal{F} \longrightarrow j_*j^*\mathcal{F}$$

To extend this to the derived category, we take an injective resolution of $\mathcal{F}$. Recall that injective sheaves are flasque. For an injective (and hence flasque) sheaf $\mathcal{I}$, the restriction map is surjective:

$$\mathcal{I}(V) \longrightarrow (j_*j^*\mathcal{I})(V) = \mathcal{I}(V \cap U) \text{ is surjective.}$$

To properly extend this to the derived category $D(X)$, we replace $\mathcal{F}$ with its injective resolution $\mathcal{I}^\bullet$. Since a sheaf and its injective resolution are quasi-isomorphic, they are isomorphic in $D(X)$.

Furthermore, because injective sheaves are flasque, the restriction map $\mathcal{I}^\bullet \longrightarrow j_*j^*\mathcal{I}^\bullet$ is surjective in every degree. This elevates our previous left-exact sequence to a short exact sequence of complexes:

$$0 \longrightarrow i_*i^!\mathcal{I}^\bullet \longrightarrow \mathcal{I}^\bullet \longrightarrow j_*j^*\mathcal{I}^\bullet \longrightarrow 0$$

A short exact sequence of complexes naturally induces a distinguished triangle in the derived category, yielding $i_*i^!\mathcal{F} \longrightarrow \mathcal{F} \longrightarrow j_*j^*\mathcal{F} \xrightarrow{+1}$. This completes the proof. □

例 1.9.1 (Relative cohomology)**.** *Let $i\colon Z \hookrightarrow X$ be a closed embedding, and let $j\colon U = X \setminus Z \hookrightarrow X$ be the open complement.*

*For a $\mathbb{C}$-vector space M, we define the **relative cohomology** of the pair (X, Z) with coefficients in M as:*

$$H^k(X, Z; M) := H^k(X; j_!M_U)$$

Applying the open-closed distinguished triangle to the constant sheaf M_X:

$$j_!j^*M_X \longrightarrow M_X \longrightarrow i_*i^*M_X$$

Notice that $j_!j^*M_X = j_!M_U$ and $i_*i^*M_X = i_*M_Z$. Taking the long exact sequence of cohomology yields the standard sequence for relative cohomology:

$$\cdots \longrightarrow H^k(X, Z; M) \longrightarrow H^k(X; M) \longrightarrow H^k(Z; M_Z) \longrightarrow \cdots$$

例 1.9.2. *Consider the compactly supported cohomology of the real line $\mathbb{R}$:*

$$H_c^k(\mathbb{R}) = \begin{cases} \mathbb{C} & k = 1 \\ 0 & k \neq 1 \end{cases}$$

This satisfies Poincaré duality when compared against the standard cohomology of $\mathbb{R}$:

$$H^k(\mathbb{R}) = \begin{cases} \mathbb{C} & k = 0 \\ 0 & k \neq 0 \end{cases}$$

Proof. The topological space $\mathbb{R}$ is homeomorphic to the open interval $(0, 1)$. Because compactly supported cohomology is a topological invariant under homeomorphisms, we immediately have:

$$H_c^k(\mathbb{R}) \cong H_c^k((0, 1))$$

First, since the closed interval $[0, 1]$ is compact, its compactly supported cohomology coincides with its standard cohomology, which in turn is isomorphic to its singular cohomology. That is,

$$H_c^k([0, 1]) = H^k([0, 1]) \cong H^k_{\text{sing}}([0, 1])$$

This yields the standard result:

$$H_c^k([0, 1]) = \begin{cases} \mathbb{C} & \text{if } k = 0, \\ 0 & \text{if } k > 0. \end{cases}$$

注 1.9.2. *In general, this isomorphism holds for locally contractible spaces. Here, the space is hereditarily paracompact.*

Second, let $X = [0,1]$. Consider the closed subset $Z = \{1\}$ with inclusion $i\colon Z \hookrightarrow X$, and the open complement $U = [0,1)$ with inclusion $j\colon U \hookrightarrow X$.

We apply the compactly supported cohomology functor H_c^k to the distinguished triangle associated with the constant sheaf $\mathbb{C}_X$:

$$j_!\mathbb{C}_U = j_!j^*\mathbb{C}_X \longrightarrow \mathbb{C}_X \longrightarrow i_*i^*\mathbb{C}_X = i_*\mathbb{C}_{\text{pt}} \xrightarrow{+1}$$

This induces a long exact sequence in compactly supported cohomology:

$$0 \longrightarrow H_c^0([0,1)) \longrightarrow H^0([0,1]) \xrightarrow{\cong} H^0(\text{pt}) \longrightarrow H_c^1([0,1)) \longrightarrow H^1([0,1]) \longrightarrow H^1(\text{pt}) \longrightarrow \cdots$$

Substituting the known cohomology groups (where $H^0([0,1]) \cong \mathbb{C}$, $H^0(\text{pt}) \cong \mathbb{C}$, and the degree 1 terms are 0), we obtain:

$$0 \longrightarrow H_c^0([0,1)) \longrightarrow \mathbb{C} \xrightarrow{\cong} \mathbb{C} \longrightarrow H_c^1([0,1)) \longrightarrow 0 \longrightarrow 0 \longrightarrow \cdots$$

Because the map between the $\mathbb{C}$ terms is an isomorphism, it forces the surrounding terms to vanish. Therefore, we conclude that:

$$H_c^k([0,1)) = 0 \quad \text{for all } k \in \mathbb{Z}.$$

Third, we repeat the process by setting $X = [0,1)$. Consider the closed subset $Z = \{0\}$ with inclusion $i\colon Z \hookrightarrow X$, and the open complement $U = (0,1)$ with inclusion $j\colon U \hookrightarrow X$.

Taking the long exact sequence in compactly supported cohomology for this new decomposition yields:

$$0 \longrightarrow H_c^0((0,1)) \longrightarrow H_c^0([0,1)) \longrightarrow H_c^0(\text{pt}) \longrightarrow H_c^1((0,1)) \longrightarrow H_c^1([0,1)) \longrightarrow H_c^1(\text{pt}) \longrightarrow \cdots$$

From our previous step, we know that $H_c^0([0,1)) = 0$ and $H_c^1([0,1)) = 0$. We also know that for a point, $H_c^0(\text{pt}) = \mathbb{C}$ and $H_c^1(\text{pt}) = 0$. Substituting these values into the exact sequence gives:

$$0 \longrightarrow H_c^0((0,1)) \longrightarrow 0 \longrightarrow \mathbb{C} \xrightarrow{\cong} H_c^1((0,1)) \longrightarrow 0 \longrightarrow 0 \longrightarrow \cdots$$

This sequence implies that $H_c^0((0,1)) = 0$ and $H_c^1((0,1)) \cong \mathbb{C}$. Consequently, we arrive at the final computation:

$$H_c^k((0,1)) = \begin{cases} \mathbb{C} & \text{if } k = 1, \\ 0 & \text{if } k \neq 1. \end{cases}$$

□

Let $\mathcal{F} \in D(X)$ and $\mathcal{G} \in D(Y)$ be objects in the derived categories of spaces X and Y, respectively. Consider the natural projection maps from the product space $X \times Y$:

$$X \xleftarrow{pr_1} X \times Y \xrightarrow{pr_2} Y$$

The **external tensor product** of $\mathcal{F}$ and $\mathcal{G}$, denoted $\mathcal{F} \boxtimes \mathcal{G} \in D(X \times Y)$, is defined as:

$$\mathcal{F} \boxtimes \mathcal{G} := pr_1^*\mathcal{F} \otimes^L pr_2^*\mathcal{G}$$

命题 1.9.1. *Let $f\colon X \longrightarrow X'$ and $g\colon Y \longrightarrow Y'$ be continuous maps. For any complexes of sheaves $\mathcal{F} \in D^+(X)$ and $\mathcal{G} \in D^+(Y)$, there is a natural isomorphism:*

$$f_!\mathcal{F} \boxtimes g_!\mathcal{G} \cong (f \times g)_!(\mathcal{F} \boxtimes \mathcal{G}).$$

Proof. The proof of this isomorphism is left as a homework exercise. □

If we apply this proposition to the case where X' and Y' are single points, we obtain the isomorphism:

$$R\Gamma_c(X, \mathcal{F}) \otimes R\Gamma_c(Y, \mathcal{G}) \cong R\Gamma_c(X \times Y, \mathcal{F} \boxtimes \mathcal{G}), \tag{1.6}$$

which is the statement of the **Künneth formula**.

As an immediate consequence, we can compute the compactly supported cohomology of Euclidean space. Since $\mathbb{R}^n$ is the product of n copies of $\mathbb{R}$, we have:

$$H_c^k(\mathbb{R}^n) = \begin{cases} \mathbb{C} & \text{if } k = n, \\ 0 & \text{if } k \neq n. \end{cases}$$

1.10 Smooth Morphisms

定义 1.10.1. *A continuous map $f\colon X \longrightarrow Y$ is called a **topological submersion** of relative dimension n if for all $x \in X$, there exists a neighborhood U of x such that $f(U)$ is open in Y, and there exists a homeomorphism $U \cong f(U) \times \mathbb{R}^n$ making the following diagram commute:*

$$\begin{array}{ccc} U & \xrightarrow{\simeq} & f(U) \times \mathbb{R}^n \\ & \searrow^{f} & \downarrow{\scriptstyle pr_1} \\ & & f(U) \end{array}$$

For such a trivializing neighborhood U, we can compute the compactly supported pushforward of the constant sheaf along the restriction $f|_U$. Let $a_{\text{pt}}\colon \mathbb{R}^n \longrightarrow \text{pt}$ denote the map to a point. We have:

$$\begin{aligned} (f|_U)_!\mathbb{C}_U &\cong (pr_1)_!\mathbb{C}_{f(U)\times\mathbb{R}^n} \\ &\cong (\text{id}_{f(U)} \times a_{\text{pt}})_!\mathbb{C}_{f(U)\times\mathbb{R}^n} \\ &\cong \mathbb{C}_{f(U)} \otimes R\Gamma_c(\mathbb{R}^n, \mathbb{C}) \\ &\cong \mathbb{C}_{f(U)} \otimes \mathbb{C}[-n] \\ &\cong \mathbb{C}_{f(U)}[-n]. \end{aligned}$$

定义 1.10.2. *Define the **relative orientation sheaf** for a topological submersion $f\colon X \longrightarrow Y$ as the sheafification of a specific presheaf:*

$$or_f := (or_{f,pre})^+$$

where the presheaf $or_{f,pre}$ evaluated on an open set U is given by:

$$or_{f,pre}(U) = \text{Hom}_{Sh(f(U))}\left(H^{-n}((f|_U)_!\mathbb{C}_U), \mathbb{C}_{f(U)}\right).$$

Over a trivializing neighborhood where $U \cong f(U) \times \mathbb{R}^n$ and $f|_U$ acts as the first projection, we can directly see that:

$$\mathrm{or}_{f|_U} \cong \mathbb{C}_U.$$

Because this property holds locally around every point, this implies that the relative orientation sheaf $\mathrm{or}_f \in \mathrm{Loc}(X)$ is a local system of rank 1 on X.

引理 1.10.1. *Let $f\colon X \longrightarrow Y$ be a topological submersion of relative dimension n. Then there is an isomorphism:*

$$f^!\mathbb{C}_Y \simeq or_f[n].$$

Proof. By definition, the relative orientation presheaf $\mathrm{or}_{f,\mathrm{pre}}$ evaluated on an open set U is given by:

$$\begin{aligned}\mathrm{or}_{f,\mathrm{pre}}(U) &= \mathrm{Hom}_{\mathrm{Sh}(f(U))}\left(H^{-n}((f|_U)_!\mathbb{C}_U), \mathbb{C}_{f(U)}\right)\\ &= \mathrm{Hom}_{\mathrm{D}(f(U))}\left((f|_U)_!\mathbb{C}_U[n], \mathbb{C}_{f(U)}\right)\end{aligned}$$

Using the adjunction between $(f|_U)_!$ and $(f|_U)^!$, we can move the pushforward to the right-hand side. Note that the adjunction takes us from the derived category over $f(U)$ to the derived category over U:

$$\begin{aligned}\mathrm{Hom}_{\mathrm{D}(f(U))}\left((f|_U)_!\mathbb{C}_U[n], \mathbb{C}_{f(U)}\right) &= \mathrm{Hom}_{\mathrm{D}(U)}\left(\mathbb{C}_U[n], (f|_U)^!\mathbb{C}_{f(U)}\right)\\ &= \mathrm{Hom}_{\mathrm{D}(U)}\left(\mathbb{C}_U, f^!\mathbb{C}_Y|_U[-n]\right).\end{aligned}$$

This perfectly matches the presheaf definition for the local sections of $f^!\mathbb{C}_Y[-n]$, which implies that the sheafification or_f is isomorphic to $f^!\mathbb{C}_Y[-n]$. Therefore, $f^!\mathbb{C}_Y \simeq \mathrm{or}_f[n]$. □

定理 1.10.1. *Let $f\colon X \longrightarrow Y$ be a topological submersion of relative dimension n as above. For any complex of sheaves $\mathcal{F} \in D^+(Y)$, there is a natural isomorphism:*

$$or_f \otimes^L f^*\mathcal{F}[n] \xrightarrow{\simeq} f^!\mathcal{F}.$$

Proof. By the preceding lemma, we have $\mathrm{or}_f \otimes^L f^*\mathcal{F}[n] \simeq f^!\mathbb{C}_Y \otimes^L f^*\mathcal{F}$. Thus, we want to construct a morphism:

$$f^!\mathbb{C}_Y \otimes^L f^*\mathcal{F} \longrightarrow f^!\mathcal{F}.$$

By the adjunction between the functors $f_!$ and $f^!$, specifying this morphism is equivalent to providing a morphism:

$$f_!\left(f^!\mathbb{C}_Y \otimes^L f^*\mathcal{F}\right) \longrightarrow \mathcal{F}.$$

Using the projection formula, the left-hand side can be rewritten as:

$$f_!\left(f^!\mathbb{C}_Y \otimes^L f^*\mathcal{F}\right) \simeq (f_!f^!\mathbb{C}_Y) \otimes^L \mathcal{F}.$$

We can now obtain the desired map to $\mathcal{F}$ by taking the standard counit map $f_!f^!\mathbb{C}_Y \longrightarrow \mathbb{C}_Y$ and tensoring it with $\mathcal{F}$.

To show that this constructed map is indeed an isomorphism, we may assume locally that $X \cong Y \times \mathbb{R}^n$ and that $f\colon Y \times \mathbb{R}^n \xrightarrow{pr_1} Y$ is the first projection. In this trivialized case, we have $\mathrm{or}_f \simeq \mathbb{C}_X$.

Checking the isomorphism locally on the cohomology sheaves $H^i(f^!\mathcal{F})$ is equivalent to evaluating the presheaf:

$$U \longmapsto \mathrm{Hom}(\mathbb{C}_U, f^!\mathcal{F}[i]).$$

We can take basic open sets of the form $U = V \times W \subseteq Y \times \mathbb{R}^n$, where $W \cong \mathbb{R}^n$ is an open ball. For such an open set, the restriction of f acts as $f|_U = \mathrm{id}_V \times a_W$, where $a_W \colon W \longrightarrow \mathrm{pt}$ is the map to a point.

Using adjunction and the Künneth formula (1.6) for pushforwards, we compute:

$$\begin{aligned}
\mathrm{Hom}(\mathbb{C}_U, f^!\mathcal{F}[i]) &\cong \mathrm{Hom}((f|_U)_!\mathbb{C}_U, \mathcal{F}[i]) \\
&\cong \mathrm{Hom}(\mathbb{C}_V \otimes a_{W!}\mathbb{C}_W, \mathcal{F}[i]) \\
&\cong \mathrm{Hom}(\mathbb{C}_V[-n], \mathcal{F}[i]) \\
&\cong \mathrm{Hom}(\mathbb{C}_V, \mathcal{F}[n][i]) \\
&\cong \mathrm{Hom}(\mathbb{C}_U, f^*\mathcal{F}[n][i]) \quad \text{because } f \text{ is surjective.}
\end{aligned}$$

Therefore, gluing these local isomorphisms together, we conclude that there is a global isomorphism:

$$\mathrm{or}_f \otimes^L f^*\mathcal{F}[n] \simeq f^!\mathcal{F}.$$

□

定理 1.10.2. *Let $f \colon X \longrightarrow Y$ be a smooth morphism of relative complex dimension d between algebraic varieties. Then the relative orientation sheaf is trivial, namely $\mathrm{or}_f \simeq \mathbb{C}_X$.*

Consequently, for any complex of sheaves $\mathcal{F} \in D^+(Y)$, there is a natural isomorphism:

$$f^*\mathcal{F}[2d] \simeq f^!\mathcal{F}.$$

Proof. The proof of this statement can be found in Achar, Proposition 2.2.8. □

注 1.10.1. *The condition of smoothness is crucially used in the proof of the following lemma.*

引理 1.10.2. *Let $U \subseteq \mathbb{C}^n$ be an open subset, and let $j \colon U \hookrightarrow \mathbb{C}^n$ be the inclusion map. Let $f \colon U \longrightarrow \mathbb{C}^n$ be a holomorphic map such that f induces a biholomorphism onto its image, i.e., $f \colon U \xrightarrow{\sim} f(U)$. Then the induced maps on compactly supported cohomology coincide and yield isomorphisms:*

$$j_\# = f_\# \colon H_c^{2n}(U) \xrightarrow{\simeq} H_c^{2n}(\mathbb{C}^n)$$

1.11 Constructibility

注 1.11.1. *Henceforth, we will assume that all spaces considered are algebraic varieties.*

Recall the definition of the constructible derived category: $D_c^b(X)$ is the full subcategory of $D^b(X)$ consisting of bounded complexes of sheaves $\mathcal{F}^\bullet$ such that for every integer k, the cohomology sheaf $\mathcal{H}^k(\mathcal{F}^\bullet)$ is a constructible sheaf.

It is an easy verification to show that $D_c^b(X)$ is a triangulated subcategory. Specifically, it is stable under the shift functor $[1]$ and under taking mapping cones.

Goal: Our primary goal in the upcoming sections is to show that the constructible derived category $D_c^b(X)$ is stable under the Grothendieck six functors.

命题 1.11.1. *Let $f\colon X \longrightarrow Y$ be a morphism between algebraic varieties. For any complex $\mathcal{F} \in D_c^b(Y)$, the pullback $f^*\mathcal{F}$ belongs to $D_c^b(X)$.*

Proof. Choose a stratification $Y = \coprod Y_t$ such that for every integer k, the cohomology sheaf restricted to the stratum, $\mathcal{H}^k(\mathcal{F})|_{Y_t}$, is a finite-rank local system (i.e., it belongs to $\mathrm{Loc}(Y_t)^{\mathrm{fin}}$).

Taking the preimages of these strata yields a stratification on X, given by $X = \coprod f^{-1}(Y_t) = \coprod X_t$. Because the pullback of a local system is a local system and the exactness of pullback, the restriction of the cohomology of the pullback, $\mathcal{H}^k(f^*\mathcal{F})|_{f^{-1}(Y_t)}$, belongs to $\mathrm{Loc}(f^{-1}(Y_t))^{\mathrm{fin}}$. This demonstrates that $f^*\mathcal{F}$ is constructible. □

命题 1.11.2. *The derived tensor product functor $\otimes^L$ maps $D_c^b(X) \times D_c^b(X)$ into $D_c^b(X)$.*

Proof. The proof is straightforward and left as an exercise. □

定理 1.11.1. *Let $f\colon X \longrightarrow Y$ be a morphism between algebraic varieties, and let $\mathcal{F} \in D_c^b(X)$. Then the pushforward complexes $f_*\mathcal{F}$ and $f_!\mathcal{F}$ both belong to $D_c^b(Y)$.*

Proof Sketch. The rigorous proof of this theorem is highly non-trivial; we refer the reader to Achar, Theorem 2.7.1. We provide a sketch of the proof below.

First, we invoke **Nagata's Compactification Theorem**: for any morphism $f\colon X \longrightarrow Y$ between algebraic varieties, there exists an algebraic variety $\tilde{X}$, an open embedding $j\colon X \hookrightarrow \tilde{X}$, and a proper morphism $\tilde{f}\colon \tilde{X} \longrightarrow Y$ such that: $f = \tilde{f} \circ j$.

Since the functors compose naturally as $f_* = \tilde{f}_* \circ j_*$ and $f_! = \tilde{f}_! \circ j_!$, it suffices to reduce the problem to verifying the statement for two specific types of morphisms: an open embedding, and a proper morphism. (Recall that for a proper morphism, $f_* = f_!$).

1. Let us consider the case where $f\colon X \hookrightarrow Y$ is an open embedding. (In fact, this assumption can be weakened to f being a locally closed embedding).
   (a) We can choose a stratification $Y = \coprod Y_s$ such that X is exactly a union of a subset of these strata, and the restriction $\mathcal{H}^k(\mathcal{F})|_{Y_s}$ belongs to $\mathrm{Loc}(Y_s)^{\mathrm{fin}}$. For the extension by zero functor $f_!$, its restriction to a stratum Y_s is given by:
   $$(f_!\mathcal{F})|_{Y_s} = \begin{cases} \mathcal{F}|_{Y_s} & \text{if } Y_s \subseteq X, \\ 0 & \text{if } Y_s \not\subseteq X. \end{cases}$$
   In both cases, the restriction is constructible. This immediately implies that $f_!\mathcal{F} \in D_c^b(Y)$.
   (b) To show that $f_*\mathcal{F}$ is constructible, we proceed through a series of reductions:
   - By using truncation distinguished triangles, we can reduce the problem from bounded complexes to considering a single sheaf $\mathcal{F}$ in the abelian category $\mathrm{Sh}(X)$.
   - We can proceed by induction on the number of irreducible components of X. By applying the open-closed distinguished triangle $j_!j^*\mathcal{F} \longrightarrow \mathcal{F} \longrightarrow i_*i^*\mathcal{F} \xrightarrow{+1}$, we can reduce the problem to the case where X is an irreducible variety.
   - Any morphism $f\colon X \longrightarrow Y$ can be factored as $X \longrightarrow \overline{f(X)} \hookrightarrow Y$, where the second map is a proper closed embedding. This allows us to reduce to the case where f is a dominant morphism.

Choose an open embedding $h\colon U \hookrightarrow X$ such that the restriction $\mathcal{L} = \mathcal{F}|_U$ is a finite-rank local system, meaning $\mathcal{L} \in \mathrm{Loc}(U)^{\mathrm{fin}}$. Let $i\colon Z = X \setminus U \hookrightarrow X$ be the closed embedding of the complement. We have the distinguished triangle:

$$i_* i^! \mathcal{F} \longrightarrow \mathcal{F} \longrightarrow h_* \mathcal{L} \xrightarrow{+1}$$

Applying the functor f_*, we obtain the distinguished triangle:

$$(f \circ i)_* i^! \mathcal{F} \longrightarrow f_* \mathcal{F} \longrightarrow f_* h_* \mathcal{L} \xrightarrow{+1}$$

To analyze the term $f_* h_* \mathcal{L}$, we need to invoke the **resolution of singularities**.

Suppose X is an irreducible variety and $Z \subsetneq X$ is a proper closed subset such that the complement $X \setminus Z$ is non-empty and smooth. There exists a proper morphism $\pi\colon \widetilde{X} \longrightarrow X$ satisfying the following properties:

i. The space $\widetilde{X}$ is a smooth variety.
ii. The restriction of π over the complement, $\pi|_{\pi^{-1}(X\setminus Z)}\colon \pi^{-1}(X \setminus Z) \longrightarrow X \setminus Z$, is an isomorphism.
iii. The preimage $\pi^{-1}(Z) \subset \widetilde{X}$ is a simple normal crossing divisor (which means locally it can be defined by an equation of the form $\prod x_i = 0$).

Using this resolution, the problem reduces to a local computation for the normal crossing case, specifically an embedding of the form $i\colon \{0\} \times \mathbb{C}^d \hookrightarrow \mathbb{C}^{n-d} \times \mathbb{C}^d$. From here, one applies an ad hoc argument (details of which can be found in the appendix of Achar's book).

2. Finally, if f is a proper morphism, one can use induction on the dimension to reduce to the case where $\dim Y = 0$. In this case, we need to show that $f_* \mathcal{F} \in D_c^b(\mathrm{pt})$, which essentially means the cohomology groups are finite-dimensional. This fact follows from the Artin Vanishing Theorem.

定理 1.11.2 (Artin Vanishing, cf. Achar Theorem 2.6.2)**.** *Let X be an affine algebraic variety, and let $\mathcal{F} \in \mathrm{Sh}_c(X)$ be a constructible sheaf. Then the complex dimension of the cohomology groups is finite, meaning $\dim_{\mathbb{C}} H^k(X, \mathcal{F}) < \infty$ for all k. Furthermore, we have:*

$$H^k(X, \mathcal{F}) = 0$$

unless $0 \leq k \leq \dim_{\mathbb{C}} \operatorname{supp} \mathcal{F} \leq \dim_{\mathbb{C}} X$.

□

命题 1.11.3. *For any complexes $\mathcal{F}, \mathcal{G} \in D_c^b(X)$, the derived sheaf Hom complex $R\mathcal{H}om(\mathcal{F}, \mathcal{G})$ belongs to $D_c^b(X)$.*

Proof. Take an open subset U with inclusion $j\colon U \hookrightarrow X$ such that the cohomology sheaves $\mathcal{H}^k(\mathcal{F}|_U)$ and $\mathcal{H}^k(\mathcal{G}|_U)$ are finite-rank local systems (i.e., they belong to $\mathrm{Loc}(U)^{\mathrm{fin}}$). Let $i\colon Z = X \setminus U \hookrightarrow X$ denote the closed embedding of the complement.

We start with the standard open-closed distinguished triangle for $\mathcal{F}$:

$$j_! j^* \mathcal{F} \longrightarrow \mathcal{F} \longrightarrow i_* i^* \mathcal{F} \xrightarrow{+1}$$

Applying the functor $R\mathcal{H}om(-,\mathcal{G})$, we obtain the following distinguished triangle:

$$R\mathcal{H}om(i_*i^*\mathcal{F},\mathcal{G}) \longrightarrow R\mathcal{H}om(\mathcal{F},\mathcal{G}) \longrightarrow R\mathcal{H}om(j_!j^*\mathcal{F},\mathcal{G}) \xrightarrow{+1}$$

By adjunction and the projection formula, we can rewrite the outer terms. The distinguished triangle becomes:

$$i_*R\mathcal{H}om(i^*\mathcal{F},i^!\mathcal{G}) \longrightarrow R\mathcal{H}om(\mathcal{F},\mathcal{G}) \longrightarrow j_*R\mathcal{H}om(\mathcal{F}|_U,\mathcal{G}|_U) \xrightarrow{+1}$$

We know that $j_*R\mathcal{H}om(\mathcal{F}|_U,\mathcal{G}|_U)$ belongs to $D_c^b(X)$ because $\mathcal{F}|_U$ and $\mathcal{G}|_U$ are local systems (making their internal Hom a local system) and the pushforward j_* preserves constructibility. Therefore, to show that the middle term $R\mathcal{H}om(\mathcal{F},\mathcal{G})$ is in $D_c^b(X)$, we only need to show that the left term $i_*R\mathcal{H}om(i^*\mathcal{F},i^!\mathcal{G})$ is in $D_c^b(X)$.

By an induction argument on the dimension of the space, it suffices to show that $i^!\mathcal{G} \in D_c^b(Z)$.

To see this, consider the standard distinguished triangle for $\mathcal{G}$:

$$i_*i^!\mathcal{G} \longrightarrow \mathcal{G} \longrightarrow j_*j^*\mathcal{G} \xrightarrow{+1}$$

By our initial assumptions, $\mathcal{G} \in D_c^b(X)$. Furthermore, because the functors j^* and j_* preserve constructibility, the term $j_*j^*\mathcal{G}$ also belongs to $D_c^b(X)$.

Since two out of the three terms in the distinguished triangle belong to the triangulated category $D_c^b(X)$, the third term must as well. This implies that $i_*i^!\mathcal{G} \in D_c^b(X)$. Finally, applying the pullback functor i^* (which also preserves constructibility), we conclude:

$$i^*(i_*i^!\mathcal{G}) \simeq i^!\mathcal{G} \in D_c^b(Z).$$

This completes the proof. □

命题 1.11.4. *Let X be an algebraic variety. Then the dualizing complex $\omega_X := a_X^!\mathbb{C}_{pt}$ belongs to $D_c^b(X)$, where $a_X\colon X \longrightarrow pt$ is the map to a point.*

In particular, the Verdier duality functor $\mathbb{D} := R\mathcal{H}om(-,\omega_X)$ is a well-defined functor from $D_c^b(X)$ to $D_c^b(X)^{op}$.

Proof. We proceed by induction on the dimension of X.

If $\dim X = 0$ (so X is a point), the statement is obvious.

For the general case, we can take an open, dense, smooth subset U with inclusion $j\colon U \hookrightarrow X$, and let $i\colon Z \hookrightarrow X$ be the closed embedding of the complement. We apply the standard open-closed distinguished triangle to the dualizing complex ω_X:

$$i_*\omega_Z = i_*i^!\omega_X \longrightarrow \omega_X \longrightarrow j_*j^*\omega_X = j_*\omega_U \xrightarrow{+1}$$

By our induction hypothesis, the term $i_*\omega_Z$ belongs to the constructible derived category $D_c^b(X)$. For the open term, because U is chosen to be a smooth variety, its dualizing complex is simply a shifted constant sheaf, $\omega_U \simeq \mathbb{C}_U[2\dim U]$. Since the pushforward along an open embedding preserves constructibility, we have $j_*\omega_U \in D_c^b(X)$.

Because two out of the three terms in the distinguished triangle belong to $D_c^b(X)$, it follows that the middle term ω_X must also belong to $D_c^b(X)$. This completes the induction and the proof. □

There is a natural evaluation map for the derived tensor product:

$$\mathbb{D}\mathcal{F} \otimes^L \mathcal{F} = R\mathcal{H}om(\mathcal{F}, \omega_X) \otimes^L \mathcal{F} \longrightarrow \omega_X.$$

By the tensor-Hom adjunction formula, we have an isomorphism of Hom-sets:

$$\mathrm{Hom}(\mathbb{D}\mathcal{F} \otimes^L \mathcal{F}, \omega_X) \cong \mathrm{Hom}(\mathcal{F}, R\mathcal{H}om(\mathbb{D}\mathcal{F}, \omega_X)) = \mathrm{Hom}(\mathcal{F}, \mathbb{D}^2\mathcal{F}).$$

The natural evaluation map on the left-hand side corresponds, under this adjunction, to a canonical evaluation morphism:

$$\mathcal{F} \longrightarrow \mathbb{D}^2\mathcal{F}.$$

命题 1.11.5. *Let X be an algebraic variety, and let $\mathcal{F} \in D_c^b(X)$. The canonical evaluation map $\mathcal{F} \xrightarrow{\simeq} \mathbb{D}^2\mathcal{F}$ is an isomorphism.*

注 1.11.2. *The condition that $\mathcal{F}$ belongs to the constructible bounded derived category $D_c^b(X)$ cannot be omitted. If one considers general complexes of sheaves, there are counterexamples even in the simplest case where X is a single point (e.g., complexes of infinite-dimensional vector spaces).*

Proof Sketch. For the case where X is a smooth variety, we have an explicit description of the dualizing complex: $\omega_X \simeq \mathbb{C}_X[2\dim X]$. In this case, the double dual is easily verified to be an isomorphism.

For a general variety X, one employs a similar trick of stratifying the space into smooth locally closed pieces and proceeding by induction on the dimension. (This process requires several other technical ingredients, which we will skip for today). For a complete proof, refer to Achar, Theorem 2.8.6. □

Recall: The Verdier duality functor interacts with the Grothendieck pushforward and pullback functors in the following ways. For any complex $\mathcal{F} \in D_c^b(X)$, there are natural isomorphisms:

$$f_*\mathbb{D}\mathcal{F} \simeq \mathbb{D}f_!\mathcal{F},$$
$$f^!\mathbb{D}\mathcal{F} \simeq \mathbb{D}f^*\mathcal{F}.$$

命题 1.11.6. *Let $f\colon X \longrightarrow Y$ be a morphism between algebraic varieties, and let $\mathcal{F} \in D_c^b(Y)$. Then the exceptional pullback $f^!\mathcal{F}$ belongs to $D_c^b(X)$.*

Proof. Using the properties of Verdier duality and the biduality isomorphism established above, we can write:

$$f^!\mathcal{F} \simeq f^!\mathbb{D}(\mathbb{D}\mathcal{F}) \simeq \mathbb{D}f^*(\mathbb{D}\mathcal{F}).$$

We already know that $\mathbb{D}$ preserves constructibility. Since $\mathcal{F} \in D_c^b(Y)$, we have $\mathbb{D}\mathcal{F} \in D_c^b(Y)$. We also know that the standard pullback f^* preserves constructibility, so $f^*(\mathbb{D}\mathcal{F}) \in D_c^b(X)$. Applying the duality functor $\mathbb{D}$ one final time, we conclude that $\mathbb{D}f^*(\mathbb{D}\mathcal{F}) \in D_c^b(X)$. □

注 1.11.3. *A similar algebraic manipulation can be applied to the proper pushforward functor. We have the sequence of isomorphisms:*

$$f_!\mathcal{F} \simeq \mathbb{D}(\mathbb{D}f_!\mathcal{F}) \simeq \mathbb{D}f_*(\mathbb{D}\mathcal{F}).$$

1.12　笔记: 表示论专题/2026.3.4

原网页

1.13　Set up for sheaf theory

There are four common settings for the study of sheaves in algebraic geometry and representation theory:

1. **Betti setting:** We consider complex analytic spaces endowed with the classical topology (not the Zariski topology).
   Note: We will exclusively focus on this setting for the duration of this course.
2. **Étale setting over $\mathbb{C}$:** We consider complex varieties equipped with the étale topology, using l-adic coefficients. This topology acts as a remedy for the topological shortcomings of the Zariski topology.
3. **Algebraic (de Rham) setting:** We consider complex varieties endowed with the Zariski topology. This setting heavily relies on the theory of $\mathcal{D}$-modules (which behave significantly differently from standard coherent sheaves).
4. **Positive characteristic setting:** We consider varieties defined over a finite field $\mathbb{F}_q$, equipped with the étale topology and using l-adic coefficients.

Relations:

$$\begin{array}{ccc} \text{Betti} & \xrightarrow{\simeq} & \text{Étale}/\mathbb{C}\ \text{(Comparison isomorphism)} \\ \simeq \big\downarrow \text{certain conditions} & & \big\downarrow \text{spread out technique} \\ \text{de Rham} & & \text{Étale}/\mathbb{F}_q \end{array}$$

- Why not use $\mathbb{C}$-sheaves on the Zariski topology?

引理 1.13.1. *Any locally constant sheaf on an irreducible topological space is constant.*

注 1.13.1. *The Zariski topology has too few open sets, making it inconvenient to use sheaves to study topological properties.*

1.14　Local Systems

定义 1.14.1. *A **local system** (of $\mathbb{C}$-vector spaces) on a topological space X is a locally constant sheaf. i.e., $\forall x \in X$, $\exists$ open $x \in U \subset X$ such that $\mathcal{F}|_U$ is constant.*

例 1.14.1. *Let $X = \mathbb{C} \setminus \{0\}$, for any $U \subset X$:*

$$\mathcal{F}(U) = \left\{ \textit{holomorphic functions } f \textit{ on } U \mid \partial_z f = \frac{1}{2z} f \right\}$$

then $\mathcal{F}$ is a local system and we can find $\mathcal{F}_x \cong \mathbb{C}z^{1/2} \quad \forall x \in X$.
But $z^{1/2}$ can't be defined on X, so $\mathcal{F}$ is not a constant sheaf.

引理 1.14.1. *Any local system on $[0,1]$ is constant.*

Proof. Let $\mathcal{L}$ be a local system on $[0,1]$ (compact). There exists a finite open covering $[0,1] = \bigcup_i U_i$ such that $\mathcal{L}|_{U_i}$ is constant. We may assume the finite open covering consists of intervals $U_i = (a_i, b_i)$ (or half-open intervals like $[0, b_i)$ and $(a_i, 1]$) such that $b_i \in U_{i+1}$. For example, let $U_1 = [0, b_1)$ and $U_2 = (a_2, b_2)$, so that their intersection is $U_1 \cap U_2 = (a_2, b_1)$, and so forth. We consider the restriction maps:

$$\mathcal{L}(U_1) \xrightarrow{\cong} \mathcal{L}(U_1 \cap U_2) \xleftarrow{\cong} \mathcal{L}(U_2).$$

This implies that the local system $\mathcal{L}$ is constant on the union $U_1 \cup U_2$. By continuing this process with induction, we conclude that $\mathcal{L}$ is constant on the entire interval $[0,1]$. □

Similarly, any local system on $[0,1] \times [0,1]$ is constant.

引理 1.14.2. *For any continuous map $f : X \to Y$ and a local system $\mathcal{L}$ on Y, the pullback $f^*\mathcal{L}$ is a local system on X.*

Proof. Locally, the pullback functor f^* sends a constant sheaf to a constant sheaf. □

1.15 Monodromy representations

Let $\mathrm{Loc}(X)$ denote the category of local systems on X.

命题 1.15.1. *For a locally connected topological space X, $\mathrm{Loc}(X)$ is an abelian category.*

Fix a base point $x_0 \in X$. Our goal is to construct a functor

$$\mathrm{Loc}(X) \longrightarrow \mathrm{Rep}(\pi_1(X, x_0))$$

where $\mathrm{Rep}(\pi_1(X, x_0))$ represents the category of finite dimensional representations $\{(V, \pi_1(X, x_0) \to \mathrm{GL}(V))\}$.

For any path $\gamma : [0,1] \to X$ and any local system $\mathcal{L} \in \mathrm{Loc}(X)$, note that the pullback $\gamma^*\mathcal{L}$ is constant on $[0,1]$ by the previous lemma. This gives us a natural isomorphism $\rho(\gamma)$ through the global sections $\Gamma([0,1], \gamma^*\mathcal{L})$. Specifically, we have the following identifications of stalks:

$$\mathcal{L}_{\gamma(1)} = (\gamma^*\mathcal{L})_1 \xleftarrow{\cong} \Gamma([0,1], \gamma^*\mathcal{L}) \xrightarrow{\cong} (\gamma^*\mathcal{L})_0 = \mathcal{L}_{\gamma(0)}.$$

For a loop where $\gamma(0) = \gamma(1) = x_0$, we obtain an isomorphism

$$\rho(\gamma) : \mathcal{L}_{x_0} \xrightarrow{\cong} \mathcal{L}_{x_0}.$$

That is, $\rho(\gamma) \in \mathrm{GL}(\mathcal{L}_{x_0})$.

引理 1.15.1. *The assignment $\rho(\gamma)$ induces a well-defined action of the fundamental group $\pi_1(X, x_0)$ on the vector space $V = \mathcal{L}_{x_0}$.*

Proof. To show that this action is well-defined, we need to verify the following properties:

1. If γ is the identity path, then $\rho(\gamma)$ is the identity isomorphism.
2. The assignment respects path composition.

3. The assignment is invariant under homotopy. That is, if two paths are homotopic ($\gamma \sim \gamma'$), then their induced isomorphisms are equal, meaning $\rho(\gamma) = \rho(\gamma')$.

To prove the homotopy invariance, let $H : [0,1] \times [0,1] \to X$ be a homotopy between γ and γ', such that $H(t,0) = \gamma(t)$ and $H(t,1) = \gamma'(t)$. Since the pullback $H^*\mathcal{L}$ is a constant local system on the square $[0,1] \times [0,1]$, its restriction to the boundary paths is also constant. This yields $\gamma^*\mathcal{L} = \gamma'^*\mathcal{L}$, which implies that $\rho(\gamma) = \rho(\gamma')$. □

We have thus defined the monodromy representation functor:

$$\begin{aligned} \mathrm{Mon}_{x_0} : \mathrm{Loc}(X) &\longrightarrow \mathrm{Rep}(\pi_1(X, x_0)) \\ \mathcal{L} &\longmapsto \mathcal{L}_{x_0}. \end{aligned}$$

例 1.15.1. *Consider the previous example of the local system associated with $z^{1/2}$ on $X = \mathbb{C} \setminus \{0\}$. Let the base point be $x_0 = 1$. The fundamental group is $\pi_1(X, x_0) \cong \mathbb{Z}$. The monodromy representation maps the generator of the fundamental group to an element in $\mathrm{GL}(\mathbb{C}) \cong \mathbb{C}^*$. Specifically, as z goes around the origin once, the branch of $z^{1/2}$ changes sign, so the generator $1 \in \mathbb{Z}$ is mapped to $-1 \in \mathbb{C}^*$ (i.e., $z^{1/2} \mapsto -z^{1/2}$).*

定理 1.15.1. *Suppose the topological space X is connected, locally path-connected, and locally simply-connected. Then the monodromy functor Mon_{x_0} is an equivalence of categories.*

注 1.15.1. *For example, topological manifolds satisfy these local conditions. In addition, cones also satisfy these local conditions.*

Fact: For the inverse functor, Let $p : \tilde{X} \to X$ be the universal covering space. The fundamental group $\pi_1(X, x_0)$ acts on $\tilde{X}$ via deck transformations on the fibers of p.

Given a representation $V \in \mathrm{Rep}(\pi_1(X, x_0))$, we define a presheaf $\mathcal{L}_V$ by assigning to each open set $U \subseteq X$:

$$\mathcal{L}_V(U) = \left\{ \begin{array}{l} \text{locally constant, } \pi_1(X,x_0)\text{-equivariant maps } \phi : p^{-1}(U) \to V \\ \text{i.e., } \phi(g \cdot y) = g \cdot \phi(y) \text{ for any } g \in \pi_1(X,x_0) \text{ and } y \in p^{-1}(U) \end{array} \right\}.$$

One can verify that $\mathcal{L}_V$ is indeed a sheaf. Furthermore, we have an isomorphism on the stalk at the base point: $(\mathcal{L}_V)_{x_0} \cong V$. And for any local system $\mathcal{L}$, we have $\mathcal{L}_{\mathcal{L}_{x_0}} \cong \mathcal{L}$. (*Hint:* Fix a point $\tilde{x}_0 \in p^{-1}(x_0)$. For any $x \in X$, fix a path α_x from x_0 to x (fixing this data ensures consistency), with the condition that the chosen path for the base point α_{x_0} is the identity path.)

引理 1.15.2. *Let X be as in the previous theorem (which implies X is path-connected). For any local system $\mathcal{L} \in \mathrm{Loc}(X)$ corresponding to a representation $V \in \mathrm{Rep}(\pi_1(X, x_0))$, there is an isomorphism of global sections:*

$$\Gamma(X, \mathcal{L}) \cong V^{\pi_1(X, x_0)}.$$

Proof. We have a natural evaluation map $\Gamma(X, \mathcal{L}) \to \mathcal{L}_{x_0} = V$.

For any loop $\gamma \in \pi_1(X, x_0)$ where $\gamma(0) = \gamma(1) = x_0$, the monodromy action is given by the composition of isomorphisms:

$$\rho(\gamma) : V = \mathcal{L}_{x_0} \xleftarrow{\cong} \Gamma([0,1], \gamma^*\mathcal{L}) \xrightarrow{\cong} \mathcal{L}_{x_0}.$$

For any global section $s \in \Gamma(X, \mathcal{L})$, its pullback along the loop yields $s|_{[0,1]} \in \Gamma([0,1], \gamma^*\mathcal{L})$. This implies that $\rho(\gamma)(s_{x_0}) = s_{x_0}$. Thus, the image of the evaluation map lies in the invariant subspace $V^{\pi_1(X,x_0)}$.

To show this map is an isomorphism, we prove injectivity and surjectivity separately:

- **Injectivity:** For any two global sections $s, s' \in \Gamma(X, \mathcal{L})$, suppose their stalks at x_0 are equal, meaning $s_{x_0} = s'_{x_0}$. For any arbitrary point $x \in X$, take a path α such that $\alpha(0) = x_0$ and $\alpha(1) = x$. This path induces an isomorphism $\rho(\alpha) : \mathcal{L}_{x_0} \xrightarrow{\cong} \mathcal{L}_x$. Because s and s' are global sections, we have $\rho(\alpha)(s_{x_0}) = s_x$ and $\rho(\alpha)(s'_{x_0}) = s'_x$. Given that $s_{x_0} = s'_{x_0}$, it follows immediately that $s_x = s'_x$. Since x is arbitrary, we conclude that $s = s'$.
- **Surjectivity:** Let $v \in V^{\pi_1(X,x_0)}$ be an invariant element. For any $x \in X$, take a path α from x_0 to x. This path induces an isomorphism $\rho(\alpha) : \mathcal{L}_{x_0} \xrightarrow{\cong} \mathcal{L}_x$. We can define the stalk at x by setting $v_x := \rho(\alpha)(v)$. Because v is invariant under the action of the fundamental group $\pi_1(X, x_0)$, this definition is independent of the choice of the path α. This process yields a well-defined global section $s \in \Gamma(X, \mathcal{L})$ such that $s_{x_0} = v$.

□

1.16 Constructible Sheaves

It is not enough to consider the category of local systems $\mathrm{Loc}(X)$ for the following reasons:

1. It only detects the fundamental group $\pi_1(X, x_0)$.
2. It is not stable under functors.

例 1.16.1. *Let $\mathcal{L}$ be the local system associated with $z^{1/2}$ on $X = \mathbb{C} \setminus \{0\}$. Consider the inclusion map $j : X \hookrightarrow Y = \mathbb{C}$, and the pushforward sheaf $j_*\mathcal{L}$ on Y. Let us compute the stalk at the origin:*

$$(j_*\mathcal{L})_0 = \varinjlim_{0 \in V \subseteq Y} (j_*\mathcal{L})(V) = \varinjlim_{0 \in V} \mathcal{L}(V \setminus \{0\}) = 0.$$

Since the stalk at 0 is zero but the stalks elsewhere are non-zero, the sheaf $j_*\mathcal{L}$ is not a local system on Y. Thus, $j_*\mathcal{L} \notin \mathrm{Loc}(Y)$.

定义 1.16.1. *A **stratification** of a topological space X is a finite collection of subspaces $(X_i)_{i \in I}$ of X, called **strata**, such that:*

1. *$X = \bigcup_{i \in I} X_i$.*
2. *$X_i \cap X_j = \emptyset$ for any $i \neq j$.*
3. *Every stratum X_i is a connected topological manifold.*
4. *Every stratum X_i is locally closed in X.*
5. *If $X_i \cap \overline{X_j} \neq \emptyset$, then $X_i \subseteq \overline{X_j}$. (This is equivalent to the frontier condition: the boundary $\partial X_i = \overline{X_i} \setminus X_i$ is a union of some strata X_j).*

注 1.16.1. *When X is an algebraic variety, we require that each stratum X_i is a smooth, connected, Zariski locally closed subvariety.*

定义 1.16.2. *A **refinement** of a stratification $X = \bigsqcup_{i \in I} X_i$ is another stratification $X = \bigsqcup_{j \in J} X_j$ such that every stratum X_j is contained in some stratum X_i.*

例 1.16.2. *1. If X is a manifold, then it has a trivial stratification $X = X_1$.*

2. Let $X = \{(x, y) \in \mathbb{C}^2 \mid xy = 0\} \subset \mathbb{C}^2$. We can stratify X as:

$$X = \{0\} \sqcup (X \setminus \{0\}) = \{0\} \sqcup (x\text{-axis} \setminus \{0\}) \sqcup (y\text{-axis} \setminus \{0\}).$$

3. Let G be an algebraic group (for example, $\mathrm{GL}_n(\mathbb{C})$) and X be an algebraic variety equipped with a G-action that has finitely many orbits. This naturally induces a stratification $X = \bigsqcup(G\text{-orbits})$. As a specific case, consider the general linear group $\mathrm{GL}_n(\mathbb{C})$ acting by conjugation on the nilpotent cone $\mathcal{N} = \{A \in M_{n\times n}(\mathbb{C}) \mid A^n = 0\}$. The set of orbits $|\mathrm{GL}_n(\mathbb{C}) \setminus \mathcal{N}|$ is in bijection with the set of partitions of n.

引理 1.16.1. *Let X be an algebraic variety.*

1. Any two stratifications of X have a common refinement.

2. For any locally closed subvariety $Y \subseteq X$, Y is a union of strata for some stratification of X.

Proof. (1) This is left as a homework exercise. The general strategy is to consider the smooth locus of the variety and proceed using Noetherian induction.

(2) The proof follows a similar idea. □

注 1.16.2. *In more general topological settings, there exist certain regularity conditions one might need to impose on a stratification, such as the Whitney stratification conditions. However, we can safely avoid these technical conditions when we are working exclusively with algebraic varieties.*

定义 1.16.3. *Let X be a topological space equipped with a stratification $(X_i)_{i\in I}$.*

*A sheaf $\mathcal{F}$ on X is **weakly constructible** with respect to I if its restriction $\mathcal{F}|_{X_i} \in \mathrm{Loc}(X_i)$ for all $i \in I$. A sheaf is simply called weakly constructible if it is weakly constructible with respect to some stratification I.*

*Furthermore, a sheaf is **constructible** (with respect to I) if, moreover, for all $i \in I$, the restriction $\mathcal{F}|_{X_i}$ is of finite rank (i.e., all of its stalks are finite-dimensional vector spaces).*

We denote the category of constructible sheaves on X (with respect to a fixed stratification I) by $\mathrm{Sh}_c(X)$ (respectively, $\mathrm{Sh}_I(X)$).

例 1.16.3. *1. If X is a smooth variety, then the category of finite-rank local systems $\mathrm{Loc}^{\mathrm{fin}}(X)$ is a subcategory of $\mathrm{Sh}_c(X)$.*

2. For the local system $\mathcal{L}$ associated with $z^{1/2}$ on $\mathbb{C} \setminus \{0\}$, let $j : \mathbb{C} \setminus \{0\} \hookrightarrow \mathbb{C}$ be the inclusion map. Then the pushforward sheaf $j_*\mathcal{L}$ belongs to $\mathrm{Sh}_c(X)$ (where $X = \mathbb{C}$).

On derived categories:

定义 1.16.4. *A complex of sheaves $\mathcal{F}^\bullet$ is called **(weakly) constructible** (with respect to I) if its k-th cohomology sheaf $\mathcal{H}^k(\mathcal{F}^\bullet)$ satisfies the same property for all $k \in \mathbb{Z}$.*

We denote the corresponding derived categories as $D^b_I(X) \subseteq D^b(X)$ and $D^b_c(X) \subseteq D^b(X)$, where $D^b(X)$ stands for the bounded derived category of sheaves on X. Both $D^b_I(X)$ and $D^b_c(X)$ form full triangulated subcategories.

There is a natural functor connecting the derived category of constructible sheaves to the constructible derived category:

$$D^b(\mathrm{Sh}_c(X)) \longrightarrow D^b_c(X).$$

定理 1.16.1. *For any complex algebraic variety X, the natural functor from the bounded derived category of constructible sheaves to the constructible derived category,*

$$D^b(\mathrm{Sh}_c(X)) \xrightarrow{\simeq} D^b_c(X),$$

is an equivalence of categories.

Proof. It can be found in Beilinson's foundational notes, specifically in the work titled "On the derived category of perverse sheaves". □

注 1.16.3. *It is important to note that this equivalence generally fails for complex analytic spaces. However, interestingly, the equivalence still holds for real analytic spaces.*